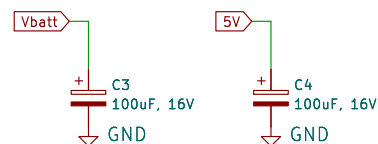
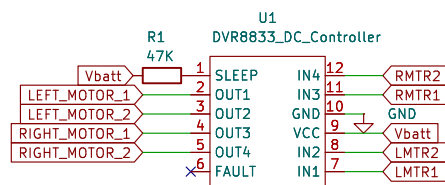


Mezzanine connections

Vcc is the output voltage from the MicroController 3.3v



Power Smoothing



M1
N20 MOTOR

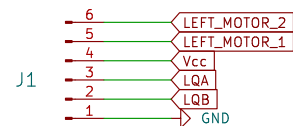


N20 MOTOR

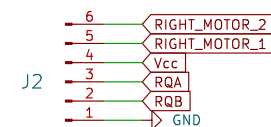
The standard build uses Pololu Item #4772 JST-SH side entry break out boards for the motor connections.

Short JST-SH to JST-SH cables are required and builders should ignore the markings on the Pololu part as these show Pololu pin sequencing whereas Gemini uses Pimoroni encoders.

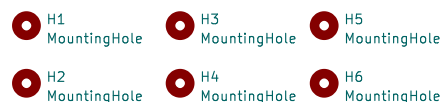
Alternative methods of connecting the motors to the chassis can be used at the builders discretion eg direct solder to pads or fitting right angle header and connecting with DuPont style connectors.



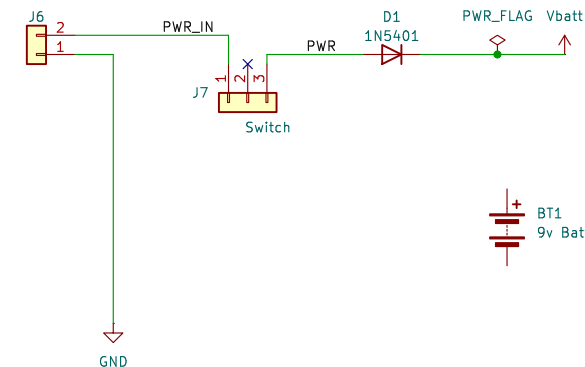
LEFT MOTOR BREAKOUT



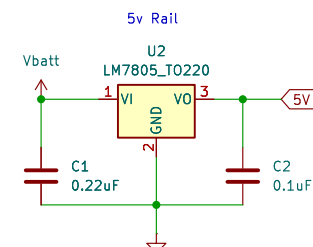
RIGHT MOTOR BREAKOUT



MECHANICAL



POWER



L4941BV can be used as a replacement if using a lipo battery
 c1 needs to be 0.1uF
 c2 needs to be 22uF

Other 5v regulators can be used in TO-220 packaging. Update cap's as required

UKMARS Gemini Chassis Board Schematic

Sheet: /

File: UKMARSBOT_RP2040Chassis.kicad_sch

Title: UKMARS Gemini Chassis 2.0

Size: A4 Date: 2025-08-14

KiCad E.D.A. 8.0.8

Rev: 2

Id: 1/1